

Solutions to First Problem Sheet

on
General Topology

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Topic: Metric Spaces

"All of mathematics is some kind of metric space."
-Israel Gelfand

Problem 1

Let $x \in \mathbb{R}^n$ and let $s = \sup\{|\pi^j(x)|\}_{j=1}^n$. Show that

$$s \leq |x| \leq s\sqrt{n}.$$

Solution 1

$$|x| = \sqrt{\sum_{i=1}^n \pi^i(x)^2} \geq \pi^j(x),$$

for any $j = 1, \dots, n$, in particular, for the supremum s , the inequality holds. Also, each component $x_i \equiv \pi^i(x)$ satisfies $|x_i| \leq s$, so $|x_i|^2 = x_i^2 \leq s^2$ so that

$$|x| \leq \sqrt{s^2 n} = s\sqrt{n},$$

and the proof is done. ☆

Problem 2

Suppose M is a metric space.

- i) Show that an open ball in M is an open subset, and a closed ball in M is a closed subset.
- ii) Show that a subset of M is open if and only if it is the union of some collection of open balls.

Solution 2

Both proofs on i) are straightforward and similar, so we only prove the second part. Given a closed ball

$$\overline{B}_r(x) = \{y \in M : d(x, y) \leq r\},$$

we take the complement C which is

$$C = \{y \in M : d(x, y) > r\}$$

and thus for any given $p \in C$, take $\delta = d(p, x) - r > 0$ so that if $b \in B_\delta(p)$, then $d(b, p) < \delta = d(p, x) - r$, whence $d(b, p)$ and then by the triangle inequality we have $d(p, x) \leq d(p, b) + d(b, x)$ so $d(p, b) \geq d(p, x) - d(b, x)$ and thus $d(p, x) - d(b, x) < d(p, x) - r$ which in turn implies that $d(b, x) > r$ so $b \in C$ and the conclusion now readily follows.

ii) ($\Leftarrow$) This is a direct consequence of the properties of open subsets on metric spaces.
 ($\Rightarrow$) Conversely, if M is open, then given any $p \in M$ let $\varepsilon_p > 0$ be such that the ball of radius ε_p and centre p is contained in M , in symbols, $B_{\varepsilon_p}(p) \subseteq M$. It is then easy to verify that the claim

$$\bigcup_{p \in M} B_{\varepsilon_p}(p) = M$$

holds true. We therefore conclude that M is the union of some collection of open balls and the proof is done. $\star$

Problem 3

Let M and N be metric spaces and let $f : M \rightarrow N$ be a map. Show that f is continuous if and only if it takes convergent sequences to convergent sequences and limits to limits, that is, if and only if $x_n \rightarrow x$ in M implies $f(x_n) \rightarrow f(x)$ in N .

Solution 3

($\Rightarrow$) If f is continuous, then take a convergent sequence $x_n \rightarrow x$. This means that given any $\gamma > 0$, there exists $N_\gamma \in \mathbb{N}$ such that

$$d(x_n, x) < \gamma \quad \text{for } n \geq N_\gamma. \quad (*)$$

Continuity of f implies that given $\varepsilon > 0$ there is some $\delta > 0$ such that $d(f(x_n), f(x)) < \varepsilon$ for $d(x_n, x) < \delta$, so that given any ε choose N_δ to satisfy (*) so that $d(x_n, x) < \delta$ for $n \geq N_\delta$ and hence $d(f(x_n), f(x)) < \varepsilon$ for $n \geq N_\delta$ and $(f(x_n))$ is a convergent sequence.

($\Leftarrow$) For the converse, we prove this with a pointwise contrapositive argument, i.e., we prove that if f is not continuous at some $x_0 \in M$, then there is some convergent sequence $x_n \rightarrow x_0$ such that $f(x_n)$ does not converge to $f(x_0)$. Suppose that f is not continuous at x_0 , this means that there is an ε_0 -ball $B_{\varepsilon_0}(f(x_0))$ such that no matter what δ -ball of x_0 we pick, there will be at least one element $x_\delta \in M \cap B_\delta(x_0)$ with $x_\delta \neq x_0$ such that $f(x_\delta) \notin B_{\varepsilon_0}(f(x_0))$. In particular, for any $n \in \mathbb{N}$, the $1/n$ -ball $B_{1/n}(x_0)$ contains some x_n such that

$$0 < |x_n - x_0| < \frac{1}{n} \quad \text{and} \quad x_n \in M,$$

but such that

$$|f(x_n) - f(x_0)| \geq \varepsilon_0 \quad \text{for all } n \in \mathbb{N}.$$

We therefore conclude that $x_n \rightarrow x_0$ but $f(x_n)$ does not converge to $f(x_0)$ and the proof is done. $\star$

Problem 4

Suppose A is a closed subset of a metric space M and (x_n) is a sequence of points in A that converges to a point $x \in M$. Show that $x \in A$.

Solution 4

Let $(x_n) \subseteq A$ with $A \subseteq M$ closed, and let $x_n \rightarrow x$. Then suppose that $x \notin A$, this means that x is in the complement of A and thus there is some $\varepsilon > 0$ such that $B_\varepsilon(x) \subseteq M \setminus A$, but if $x_n \rightarrow x$, then for such an ε we can find $N \in \mathbb{N}$ such that $d(x_n, x) < \varepsilon$ for $n \geq N$, i.e., $x_n \in B_\varepsilon(x)$ for $n \geq N$, clearly, a contradiction since $x_n \in A$ for all n , and thus the proof is done. $\star$

Problem 5

Prove that every convergent sequence in a metric space is Cauchy, and every Cauchy sequence is bounded.

Solution 5

Let (x_n) be convergent, say to x , then for $\varepsilon = 1$ we have that there is some N_1 such that $d(x_n, x) < 1$ for $n \geq N_1$, thus the triangle inequality implied that $d(x_n, 0) \leq d(x_n, x) + d(x, 0)$ so letting $d(x, 0) = D$, we get that $d(x_n, 0)$ is bounded by $1 + D$ for $n \geq N_1$, so defining $S = \sup\{d(x_i, 0)\}_{i < N_1}$ we get that the sequence is bounded by $\sup\{S, 1 + D\}$. $\star$

Problem 6

Prove the open subset criterion for continuity.

Solution 6

The statement is:

A map $f : M_1 \rightarrow M_2$ between metric spaces is continuous if and only if the preimage of every open subset is open.

($\Rightarrow$) If f is continuous, take U to be open in M_2 , then for any $p \in f^{-1}(U)$ we have that $f(p) \in U$, so there is some $\varepsilon > 0$ such that $B_\varepsilon(f(p)) \subseteq U$ and since clearly $p \in f^{-1}(B_\varepsilon(f(p)))$, then by continuity of f , there is some $\delta > 0$ for which $B_\delta(p) \subseteq f^{-1}(B_\varepsilon(f(p)))$ and using that $B_\varepsilon(f(p)) \subseteq U$ we conclude that $B_\delta(p) \subseteq f^{-1}(U)$ so $f^{-1}(U)$ is open.

($\Leftarrow$) Conversely, assume that the preimage of opens is open, then given any $\varepsilon > 0$ and any $x_0 \in M_1$, $B_\varepsilon(f(x_0))$ is open so $f^{-1}(B_\varepsilon(f(x_0)))$ is open by assumption, and it is obvious that $x_0 \in f^{-1}(B_\varepsilon(f(x_0)))$, which implies –by openness– that there is some $\delta > 0$ such that $B_\delta(x_0) \subseteq f^{-1}(B_\varepsilon(f(x_0)))$ so for any $x \in B_\delta(x_0)$ we have $f(x) \in B_\varepsilon(f(x_0))$, which means that f is continuous and the proof is complete. $\star$